

vexorin risk register

VEXORIN.AI — RISK REGISTER

Categorized Exposure Analysis with Mitigation Status · v1.0 · May 2026

Purpose. This document inventories the risks Vexorin faces, organizes them by category, and documents the mitigation status and owner for each. It is the operational artifact a Tier-1 partner reviews when assessing how systematically the team has thought about exposure.

Audience. Senior partners and analysts at Tier-1 venture firms, and the team's CFO and board for ongoing review. The document is calibrated to be re-read quarterly with mitigation status updates.

Honesty principle. Every risk in this register is real. The team's posture is to surface risks explicitly with documented mitigation paths rather than to under-disclose. A risk that surfaces in diligence and is not in this register damages trust; a risk that surfaces in diligence and is documented here demonstrates operational maturity.

Severity rating. 🚫 **KILLER** — would block a Tier-1 investment decision until resolved. 🌪️ **TERMS-AFFECTING** — would surface in IC and reduce valuation or trigger a side letter. 🤔 **DILIGENCE-QUESTION** — would surface in diligence and require a clean answer. 🟡 **LOW** — known but not material at this stage.

1. EXECUTIVE SUMMARY

The Risk Register identifies twenty-three specific risks across six categories. Of these:

- **Three are KILLER-rated** with active mitigation underway (N=1 detection track record, in-sample validation methodology, CFO underwriting credibility).
- **Six are TERMS-AFFECTING** with documented mitigation status.
- **Eleven are DILIGENCE-QUESTION** rated, expected to surface in diligence calls and pre-answered in the FAQ for Known Investor Objections.
- **Three are LOW** — operational risks tracked but not material at this stage.

No risks in this register are unmitigated and unaddressed. The team posture is to convert each KILLER and TERMS-AFFECTING item into a DILIGENCE-QUESTION-rated item by Series A through documented execution against stated mitigation plans.

2. TECHNICAL RISKS

TR-1 🚫 N=1 real-time detection track record

Description. Vexorin has one auditable real-time detection (Iran 2026). The 50-case library is retrospective reconstruction. A senior partner asks: "How do you defend a category-defining claim from one data point?" **Mitigation status.** Active. Three-part structural defense: (1) the macro validation (SEC enforcement up 67% YoY; the iceberg framing) demonstrates the category is real and growing; (2) the architectural validation (literature-derived weights producing seven geometrically distinct families on retrospective application) demonstrates the substrate works; (3) the round funds the buildout that takes the architecture into deployed customer environments where the next detections compound. **Owner.** Alistair Ewart (Co-CEO). **Resolution path.** Customer deployment cohort accumulates real-time detections through Year 1; Stream 3 marketplace creates additional surface area for validation; Series A conversation anchors on deployed traction rather than retrospective reconstruction. **Pre-answered in.** FAQ Q2.1.

TR-2 🤔 In-sample validation methodology

Description. The seven manipulation families were derived from clustering the 50-case library, then validated by reporting cosine similarity statistics computed on the same library. Test set equals training set; methodologically a real limit. **Mitigation status.** Active. 5-fold cross-validation analysis underway in May 2026 post-prompt-finalization window. Pre-registered acceptable thresholds: within-family cosine ≥ 0.75 out-of-sample, between-family ≤ 0.55 out-of-sample. Effect-size compression of 30-60% applied in the cross-validation pass. Output: Prior Weight Matrix v1.1 with cross-validated and compression-adjusted statistics replacing the in-sample numbers as the primary reference. **Owner.**

Engineering team via external analytical support. **Resolution path.** Cross-validation completes May/June 2026; Prior Weight Matrix v1.1 ships with out-of-sample numbers; deck slide 7 updates to cite cross-validated statistics. **Pre-answered in.** FAQ Q2.2.

TR-3 🎯 Conviction-math bounds derivation

Description. Pattern_factor [0.5, 2.0] and cascade_factor [1.0, 1.5] are research-derived heuristics, not empirically derived bounds. A research-trained partner asks where they come from. **Mitigation status.** Documented as scaffolding bounds informed by the 50-case library distribution; CALIBRATE refines them on customer-confirmed data over the first 60-90 days of deployment. The honest framing converts the question into a CALIBRATE selling point. **Owner.** CTO; engineering team. **Resolution path.** Empirical derivation of bounds (95th percentile of pattern-match scores observed in the library defines upper bound; 5th percentile defines lower bound) is included in the Days 7-30 analytical pass alongside cross-validation. Output: methodology note added to BRAIN Litepaper Section 5. **Pre-answered in.** FAQ Q2.3.

TR-4 🎯 Research foundation vs working code

Description. 49,000 words of citation provenance establishes architectural rigor but a partner doing technical diligence asks for working code. **Mitigation status.** Resolved at the architectural level. Scout and Carlton are production-ready and deploy on-prem in 2-3 days. Technical diligence is supported by a 45-minute Scout walkthrough — data input pipeline, signal generator, YAML config, audit-trail emitter — offered as standard element of first technical-diligence call. **Owner.** CTO. **Resolution path.** Scout walkthrough script and demo environment ready before first technical-diligence call. The clinching frame: “If we were generating signals at scale, we would not be at pre-seed.” **Pre-answered in.** FAQ Q2.4.

TR-5 🎯 CALIBRATE without ground-truth manipulation events

Description. Most pre-seed customers will see zero confirmed manipulation events in their first quarter. The 60-90 day customer-specific posterior promise is fragile under realistic customer-event rates. **Mitigation status.** Resolved at the architectural level. CALIBRATE refines on three feedback signals — analyst-rejected alerts (plentiful), analyst-confirmed alerts at any conviction tier, and AXIOM regime-state classifications. The 60-90 day window covers customer-specific FPR recalibration; full posterior weight refinement on confirmed events runs over 12+ months as enforcement-confirmed cases accumulate. **Owner.** CTO; engineering team. **Resolution path.** BRAIN Litepaper Section 9 documents the staged refinement; rarity-as-symptom framing reinforces that the slow accumulation of confirmed events is the structural condition Vexorin exists to address, not a methodology limit. **Pre-answered in.** FAQ Q2.5.

TR-6 🎯 Effect-size compression on deployed weights

Description. Foundation papers show 30-60% post-publication compression on most signal types. The Prior Weight Matrix derives weights from these papers; without compression applied, customers would encounter optimistic published magnitudes that don't reflect 2026-era market conditions. **Mitigation status.** Resolved at the architectural level. Effect-size compression of 30-60% is applied to every weight at deployment. Per-Agent Research Foundation Index documents compression factors per agent. Cross-validation analysis (TR-2) is run with compression applied; reported numbers reflect compression-adjusted weights. **Owner.** Engineering team. **Resolution path.** Compression policy documented in Prior Weight Matrix Part 6.2; visible in the moat slide as the disclosure that converts a hidden risk into an architectural choice. **Pre-answered in.** FAQ Q2.6.

TR-7 🎯 Simulation Agent has not been built yet

Description. The Simulation Agent that opens the Validated Data Marketplace is a 4-5 month engineering deliverable, not yet operational. Stream 3 revenue depends on it. A partner asks what happens if the build takes longer. **Mitigation status.** CTO-underwritten estimate (CTO has multiple prior exits and operating credibility). Engineering parallel streams begin Month 1 of post-close: BRAIN aggregation, ECHO/ATLAS/CIPHER, Simulation Agent design. Slip risk is bounded by the four-engineer parallel-staffed approach. Worst case (6 months instead of 4-5) shifts marketplace launch to Month 8; Year 1 revenue is more heavily front-loaded on Streams 1 and 2; Year 3 base case stays at ~\$5.3M because marketplace has time to mature. **Owner.** CTO. **Resolution path.** Six-Month Operating Plan documents weekly milestones for the Simulation Agent build with slip-detection criteria.

TR-8 🎯 Federated Registry has no third-party developers yet

Description. The Federated Registry is architecturally specified but has no published third-party signal modules at pre-seed close. Network effects in the registry compound only after first-developer publication. **Mitigation status.** Low priority for Year 1. The Federated Registry is a Series A-and-beyond layer per BRAIN Litepaper Section 20. Year 1 revenue does not depend on it; it is a structural moat for Year 2-3 expansion. **Owner.** Strategic — Alistair (Co-CEO). **Resolution path.** First third-party developer recruitment begins post-Series A. Registry governance model documented

3. COMMERCIAL RISKS

CR-1 ● CFO underwriting credibility for Year 3 revenue projection

Description. The Year 3 base case (\$5.3M revenue at 12 active customers, 2.5% marketplace capture) is the planning anchor for the round. A partner asks who underwrites this number. **Mitigation status.** CFO with forty years of institutional fundraising experience validates the trajectory at every pitch. The number is not a marketing forecast; it is a credentialed financial professional's underwritten estimate. Three-scenario projection (1% / 2.5% / 5% capture) frames the base case as the planning anchor with conservative as the floor and strong as the upside. **Owner.** CFO. **Resolution path.** CFO co-presents at every Series A first-call meeting; underwrites the trajectory in person. Financial Model with Sensitivity Analysis (companion document, CFO-owned) extends this into five-year sensitivity scenarios. **Pre-answered in.** FAQ Q3.2.

CR-2 ● Customer acquisition velocity for the underserved tier

Description. Initial GTM targets mid-AUM funds, regional asset managers, emerging managers, family offices. Velocity is faster than Tier-1 institutions but unproven at scale for Vexorin specifically. **Mitigation status.** Three founder-owned distribution paths (Amir / Peter / Alistair) executing in parallel from Day 1. LOIs in pipeline today. The first three customer deployments (Months 1-2) are the first verifiable velocity data point. **Owner.** Amir Zada (lead, Tier 1 GTM); Peter (Tier 2); Alistair (Tier 3). **Resolution path.** Customer Acquisition Strategy companion document tracks conversion metrics per tier; quarterly review with the board.

CR-3 ● Pricing model flexibility under buyer pushback

Description. Three-stream pricing (\$50K platform fee + usage-based agent consumption + marketplace economics) is calibrated to underserved-tier buyers. Some buyers will push back on the platform fee or the usage rates. **Mitigation status.** Pricing has volume discounts and custom enterprise terms negotiable per customer agreement. CFO underwrites pricing at quarterly review; pricing model refinements based on first-cohort feedback are expected within 90 days of first deployment. **Owner.** CFO; Amir (commercial lead). **Resolution path.** Pricing changes documented in Master Reference v2.x; first cohort feedback drives refinement before the underserved-tier expansion accelerates.

CR-4 ● Competitive response from Bloomberg, NICE Actimize, Behavox

Description. Established competitors with multi-billion revenue bases could pivot into multi-surface manipulation recognition. **Mitigation status.** Single-surface architectures cannot pivot to multi-surface manipulation recognition without rebuilding from scratch (3-4 years and \$10M+ in engineering investment per the deck moat slide). By that time, Vexorin's first cohort has compounded customer-validated weight intelligence that does not transfer. The falsifiability record cannot be backdated. The competitive landscape sits in seven distinct categories per the Competitive Landscape Analysis; none of them produces what Vexorin produces. **Owner.** Strategic — Alistair (Co-CEO). **Resolution path.** Competitive monitoring quarterly; competitive moat strengthens through customer deployment compound effects. **Pre-answered in.** FAQ Q3.4.

CR-5 ● In-house build threat at Tier-1 hedge funds

Description. Citadel, Millennium, Two Sigma, Renaissance, P72 have internal teams capable of building similar systems. **Mitigation status.** Two structural reasons they don't productize: (1) productized architecture is a different engineering problem than internal-deployment architecture; (2) strategic incentive to productize is weak because productizing reduces relative competitive advantage. Tier-1 internal builds are not Vexorin's competitor — they are eventual customers as the product matures into the long-term wedge. **Owner.** Strategic — Alistair (Co-CEO). **Resolution path.** Tier-1 customer engagement strategy (Year 3+) treats in-house teams as procurement gateways, not competitors. **Pre-answered in.** FAQ Q3.5.

CR-6 ● Marketplace network effect dependency

Description. Stream 3 (Validated Data Marketplace) economics scale with provider onboarding rate; provider onboarding scales with customer base; customer base scales with revenue traction. Network effect requires concurrent growth across all three. **Mitigation status.** Three-scenario projection (1% / 2.5% / 5% capture) covers the realistic range. Conservative case (1% capture) still produces ~\$450K annual marketplace revenue, demonstrating economic viability at low capture rates. The marketplace solves a real problem (data providers cannot prove signal isolation) that has no competing solution; capture risk is bounded by the addressable problem size. **Owner.** CFO; Amir (commercial lead). **Resolution path.** First five data provider onboardings in Months 6-9 establish the operational pattern; capture

rate becomes empirically anchored by Month 12.

4. REGULATORY RISKS

RR-1 ● EU AI Act high-risk classification ambiguity

Description. While Vexorin's manipulation detection use case is not classified as high-risk under Annex III (per FAQ Q4.1), regulatory interpretation evolves. The European Commission could issue guidance that pulls financial-market manipulation detection into the high-risk regime. **Mitigation status.** The architecture is built to support high-risk obligations (Article 12 audit trail, Article 13 explainability, Article 14 human oversight) regardless of classification. If the regulatory regime shifts, Vexorin satisfies the obligations natively rather than retrofitting. **Owner.** CFO; counsel. **Resolution path.** Quarterly regulatory monitoring; regulatory-tracking infrastructure engaged Days 90-180 post-close.

RR-2 ● GDPR enforcement on public-data ingestion

Description. GDPR enforcement has trended toward stricter interpretation of "legitimate interest" basis for public-data processing. Adverse case law could constrain Scout's ingestion model. **Mitigation status.** Dataminr precedent demonstrates the model has scaled to global Tier-1 financial-institutional customers under GDPR. The compliance posture mirrors Dataminr's; the regulatory exposure is shared, not novel. Right-to-be-forgotten architecture (deletion of public posts respected on next ingestion cycle) is built in. **Owner.** CFO; counsel. **Resolution path.** Counsel review of Scout's ingestion model before first European customer deployment; adjustments made in deployment configuration as regulatory guidance evolves. **Pre-answered in.** FAQ Q4.2.

RR-3 ● Cross-border data flow constraints

Description. Some customer jurisdictions impose strict data residency requirements that limit operational flexibility. **Mitigation status.** On-prem deployment architecture eliminates cross-border data flow for customer data — it stays in the customer's environment. Air-gapped deployment model for the strictest customers (sovereign wealth funds, regulated G-SIBs) eliminates the issue entirely. EU regional infrastructure planned for European-customer support. **Owner.** CTO; engineering team. **Resolution path.** Deployment-mode selection per customer at contract initiation; cross-border data flow positions documented in customer contracts.

RR-4 ● DORA compliance for European customers

Description. DORA imposes ICT third-party risk management obligations on financial-entity customers. Vexorin is exposed as a third-party ICT service provider. **Mitigation status.** Customer contracts include DORA-standard clauses (audit rights, incident reporting, sub-contracting controls, exit strategies). Digital Omnibus harmonization (proposed November 2025) will streamline overlapping AI Act / GDPR / NIS2 / DORA / Data Act framework. **Owner.** CFO; counsel. **Resolution path.** Contract templates reviewed and updated quarterly; Digital Omnibus implementation tracked.

RR-5 ● US state privacy law fragmentation

Description. California (CCPA/CPRA), Virginia (VCDPA), Colorado (CPA), Connecticut (CTDPA), and growing state privacy laws each impose different obligations. **Mitigation status.** Architectural choices (public-data-only ingestion, on-prem customer data handling) constrain exposure. The compliance burden is bounded by the data-handling architecture, not by specific state law nuances. **Owner.** CFO; counsel. **Resolution path.** State law tracking; per-state assessment as customer base expands geographically.

5. TEAM & EXECUTION RISKS

XR-1 ● CTO-led engineering execution against 4-5 month parallel build

Description. The Simulation Agent build, BRAIN aggregation engine, and three additional agents (ECHO, ATLAS, CIPHER) are concurrent engineering streams. Slip in any one delays the others. **Mitigation status.** CTO underwrites the timeline. Multiple prior exits and operating credibility. Two Tier-1 agentic advisors and a sophisticated agentic-engineer CTO reduce execution risk relative to a single-engineer plan. **Owner.** CTO. **Resolution path.** Six-Month Operating Plan documents weekly milestones with slip-detection criteria; weekly CTO standup; monthly CFO review of burn rate against milestone progress.

XR-2 ● Founder bandwidth across three GTM tiers

Description. Three founder-owned distribution paths (Amir / Peter / Alistair) require sustained attention from each

founder. Founder burnout or bandwidth conflict could slow GTM velocity. **Mitigation status.** Each path corresponds to a credential the owning founder already has — execution leverages existing institutional history rather than requiring founder time-investment in new skills. Distribution paths compound (Amir's direct sales meet a fund whose analyst is already familiar with Vexorin from Peter's LLM term optimization and Alistair's organic distribution). **Owner.** All three founders. **Resolution path.** Quarterly founder review; explicit time-allocation tracking; bandwidth bottlenecks surface in monthly board check-ins.

XR-3 ● Key-person risk on KOAT.AI substrate continuity

Description. Scout's social-layer substrate is the productionized continuation of KOAT.AI. The IP and twenty years of virality-algorithm operating history transfer with Alistair (Co-CEO). **Mitigation status.** IP assignment agreement executed (CFO-owned legal documentation set). Substrate codebase fully documented and replicable by the engineering team. Standard four-year vesting with one-year cliff. **Owner.** CFO; Alistair. **Resolution path.** Standard founder-vesting and IP-assignment compliance; documented in cap table.

XR-4 ● Advisor relationships and warm-intro path durability

Description. Engineering Advisor's direct connect into Founders Fund is the warm-intro path for Tier-1 outreach. Disruption to that relationship — advisor role change, retirement, etc. — could close the path. **Mitigation status.** Multiple Tier-1 paths in development (a16z American Dynamism, Sequoia, specialized fintech VCs) reduce single-path dependency. Engineering Advisor role at Kraken is stable; relationship with Founders Fund is established and not contingent on a single contact. **Owner.** Strategic — Alistair (Co-CEO). **Resolution path.** Multi-path Tier-1 engagement strategy documented in Customer Acquisition Strategy companion document.

6. MARKET RISKS

MR-1 ● Pre-seed category mismatch with generalist Tier-1 firms

Description. Surveillance and regtech are not categories where Founders Fund / a16z / Sequoia hunt at pre-seed. Their pre-seed practices skew toward different category fits. **Mitigation status.** Primary path runs through specialized fintech/regtech pre-seed VCs (Susa, Bain Capital fintech, Inspired, NEA fintech, Nasdaq Ventures, Citi Ventures, etc.). Tier-1 engagement runs through warm-intro path with thesis-fit partners (Trae Stephens at Founders Fund for the intelligence-community thesis). Cap-table-construction frame: pre-seed builds the foundation cap table; Series A brings the Tier-1 lead. **Owner.** Alistair (Co-CEO); CFO. **Resolution path.** Specialized VC primary path runs in parallel with Tier-1 warm-intro path; conversion at either path closes the round. **Pre-answered in.** FAQ Q6.1.

MR-2 ● Macro economic exposure to institutional surveillance budget cycles

Description. Institutional surveillance budgets can compress in downturns. A 2026-2027 economic contraction could reduce customer acquisition velocity. **Mitigation status.** Underserved-tier GTM is more resilient to budget compression than Tier-1 GTM because the buyers do not have a 20-40 analyst surveillance team to defend. Vexorin replaces capacity that buyers cannot afford to build internally — the value proposition strengthens, not weakens, in budget-compressed environments. **Owner.** CFO; Amir (commercial lead). **Resolution path.** Quarterly macro tracking; sales messaging adapts to budget environment.

MR-3 ● SEC enforcement deceleration

Description. SEC enforcement growth (67% YoY in FY2024) is the macro frame for Vexorin's market thesis. A future deceleration could undermine the iceberg framing. **Mitigation status.** The structural argument (multi-surface manipulation is invisible to single-surface surveillance) does not depend on SEC enforcement growth specifically. Even at flat enforcement levels, the surveillance-gap thesis holds because the gap is architectural. Enforcement growth is supportive evidence, not the load-bearing claim. **Owner.** Strategic — Alistair (Co-CEO). **Resolution path.** Enforcement data tracked quarterly; sales messaging adapts to current enforcement environment.

7. RISK GOVERNANCE & REVIEW

Review cadence

Risk Register reviewed quarterly by the founder team and CFO. Updates to mitigation status, addition of new risks, and re-rating of existing risks documented in version control. Annual review with the board including external advisor input on emerging risks.

Risk-disclosure principle

The team’s posture is to disclose risks explicitly. A risk that surfaces in diligence and is documented here demonstrates operational maturity; a risk that surfaces in diligence and is not in the register damages trust. This document is a working instrument, not a marketing artifact — risks are added as they are identified, regardless of whether they make Vexorin look better or worse.

Companion documents

- **FAQ for Known Investor Objections** — pre-answered questions for diligence calls
- **Data Strategy & Compliance Position** — regulatory landscape with sourced precedent
- **Six-Month Operating Plan** — milestone tracking and slip-detection (in build)
- **Financial Model with Sensitivity Analysis** — financial scenarios anchored to risk projections (CFO-owned, in build)

8. VERSION CONTROL

Version	Date	Changes
1.0	2026-05-07	Initial Risk Register. Twenty-three risks across six categories (technical, commercial, regulatory, team, market, governance). Three KILLER, six TERMS-AFFECTING, eleven DILIGENCE-QUESTION, three LOW. All risks paired with mitigation status, owner, and resolution path.